

Scaling to 500-5000 Tasks: Results as Exact Values

Quantum-inspired MRFO / DMO for cloud task scheduling · makespan in seconds · requested scaling study

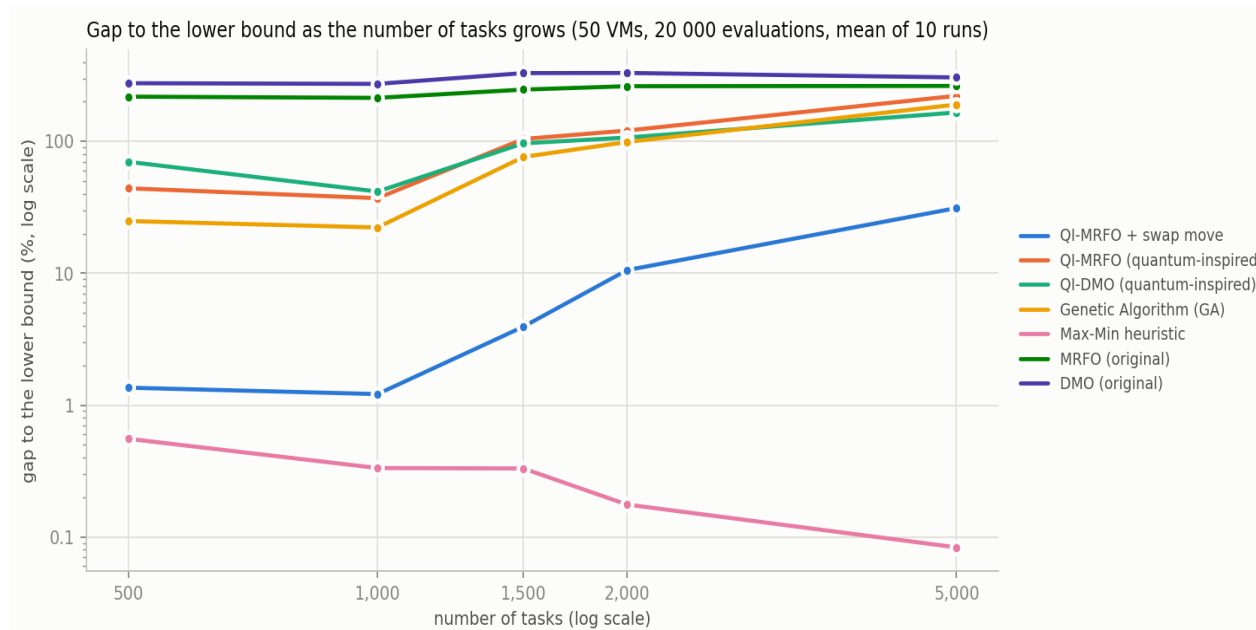
Setup. One problem per size with 500, 1,000, 1,500, 2,000, 5,000 tasks on the same kind of VM pool: 50 heterogeneous VMs (speeds 250–2000 MIPS), task lengths uniform 1000–10000 MI. Each algorithm ran 10 independent times (the deterministic heuristics once) with 20 000 evaluations per run and population 30, using the settings of the earlier studies without re-tuning (research_plan_v5.md §29).

Units. Makespan, lower bound and standard deviation: **seconds (s)** (task length in MI ÷ VM speed in MIPS). Gap: % above the lower bound, the theoretical minimum makespan. Runtime: **seconds (s)** of computer time per run.

Key results.

- **Quantum-inspired vs original** (in every run at every size for MRFO, in every run at every size for DMO): at 500 tasks MRFO 169.84 s → QI-MRFO 76.89 s and DMO 200.61 s → QI-DMO 90.73 s; at 5,000 tasks MRFO 1917.67 s → 1698.02 s and DMO 2140.14 s → 1402.89 s.
- **With the swap move, QI-MRFO beats the Genetic Algorithm** in every run at every size: 54.10 s vs 66.65 s at 500 tasks and 693.00 s vs 1532.38 s at 5,000 tasks (and GA with the same swap move in every run at every size).
- **Seeded with Max-Min, QI-MRFO + swap** is within 0.03–0.22 % of the theoretical minimum at every size and beats the Max-Min heuristic in every run at every size.

Figure 1 · Gap to the lower bound vs number of tasks



Mean of 10 runs per point (heuristic: one deterministic run). The Max-Min-seeded variants are not drawn: they stay within 0.22 % of the bound at every size (Table 2). Source: results/scale_tasks/records.csv.

Table 1 · Makespan in seconds (s): mean $\pm$ standard deviation over 10 runs, best run below

Algorithm (makespan in s)	500 tasks 50 VMs	1,000 tasks 50 VMs	1,500 tasks 50 VMs	2,000 tasks 50 VMs	5,000 tasks 50 VMs
DMO (original)	200.61 $\pm$ 4.77 best 192.25	377.84 $\pm$ 24.78 best 324.71	707.12 $\pm$ 25.89 best 644.18	995.60 $\pm$ 40.91 best 929.71	2140.14 $\pm$ 38.46 best 2096.30
QI-DMO (quantum-inspired)	90.73 $\pm$ 7.90 best 78.26	143.55 $\pm$ 4.94 best 136.98	324.00 $\pm$ 8.92 best 305.68	478.84 $\pm$ 9.57 best 463.75	1402.89 $\pm$ 26.93 best 1378.55
MRFO (original)	169.84 $\pm$ 8.26 best 153.99	317.91 $\pm$ 16.32 best 286.78	571.56 $\pm$ 43.90 best 492.11	837.58 $\pm$ 38.77 best 784.13	1917.67 $\pm$ 61.11 best 1804.64
QI-MRFO (quantum-inspired)	76.89 $\pm$ 2.35 best 71.65	138.95 $\pm$ 4.09 best 133.11	336.44 $\pm$ 21.29 best 313.86	510.95 $\pm$ 23.74 best 474.73	1698.02 $\pm$ 30.15 best 1652.69
QI-MRFO + swap move	54.10 $\pm$ 0.29 best 53.79	102.60 $\pm$ 0.77 best 101.73	171.27 $\pm$ 2.32 best 168.34	255.97 $\pm$ 3.97 best 251.19	693.00 $\pm$ 8.62 best 678.20
QI-MRFO + swap + Max-Min seed	53.49 $\pm$ 0.02 best 53.47	101.47 $\pm$ 0.01 best 101.45	164.93 $\pm$ 0.02 best 164.91	231.69 $\pm$ 0.01 best 231.67	528.30 $\pm$ 0.02 best 528.26
Genetic Algorithm (GA)	66.65 $\pm$ 3.39 best 62.86	123.90 $\pm$ 3.60 best 118.60	290.29 $\pm$ 17.18 best 252.14	460.46 $\pm$ 22.86 best 425.83	1532.38 $\pm$ 44.01 best 1476.83
GA + swap move	59.87 $\pm$ 0.84 best 59.16	108.86 $\pm$ 1.74 best 106.99	185.05 $\pm$ 5.92 best 177.26	286.70 $\pm$ 8.51 best 276.23	861.06 $\pm$ 27.32 best 812.36
PSO	147.84 $\pm$ 14.22 best 113.96	282.72 $\pm$ 12.97 best 258.10	506.36 $\pm$ 29.04 best 468.79	705.28 $\pm$ 26.59 best 661.63	1916.66 $\pm$ 50.70 best 1845.63
Random search	207.84 $\pm$ 7.38 best 197.71	362.40 $\pm$ 25.32 best 302.25	665.36 $\pm$ 18.84 best 632.68	879.13 $\pm$ 11.42 best 863.88	2296.05 $\pm$ 30.41 best 2250.33
(1+1)-EA + swap move	53.46 $\pm$ 0.03 best 53.42	101.46 $\pm$ 0.06 best 101.41	165.10 $\pm$ 0.34 best 164.87	232.64 $\pm$ 1.13 best 231.79	589.93 $\pm$ 11.36 best 575.98
(1+1)-EA + swap + Max-Min seed	53.43 $\pm$ 0.01 best 53.42	101.42 $\pm$ 0.01 best 101.41	164.83 $\pm$ 0.01 best 164.82	231.59 $\pm$ 0.01 best 231.58	528.22 $\pm$ 0.01 best 528.21
Max-Min heuristic	53.67 deterministic	101.71 deterministic	165.33 deterministic	231.96 deterministic	528.61 deterministic
Min-Min heuristic	58.64 deterministic	106.09 deterministic	170.15 deterministic	236.64 deterministic	533.16 deterministic
Lower bound (s)	53.38	101.38	164.78	231.55	528.17

Lowest mean makespan per size: 500 tasks: (1+1)-EA + swap + Max-Min seed; 1,000 tasks: (1+1)-EA + swap + Max-Min seed; 1,500 tasks: (1+1)-EA + swap + Max-Min seed; 2,000 tasks: (1+1)-EA + swap + Max-Min seed; 5,000 tasks: (1+1)-EA + swap + Max-Min seed. Source: results/scale_tasks/records.csv (analysis: results/scale_tasks_analysis.md).

Table 2 · Gap to the lower bound (%), mean over 10 runs

Algorithm (gap in %)	500 tasks	1,000 tasks	1,500 tasks	2,000 tasks	5,000 tasks
DMO (original)	275.84	272.72	329.13	329.98	305.20
QI-DMO (quantum-inspired)	69.98	41.60	96.63	106.80	165.61
MRFO (original)	218.20	213.60	246.86	261.73	263.08
QI-MRFO (quantum-inspired)	44.05	37.07	104.17	120.67	221.49
QI-MRFO + swap move	1.36	1.21	3.94	10.55	31.21
QI-MRFO + swap + Max-Min seed	0.22	0.10	0.09	0.06	0.03
Genetic Algorithm (GA)	24.87	22.22	76.17	98.86	190.13
GA + swap move	12.16	7.38	12.30	23.82	63.03
PSO	176.98	178.89	207.29	204.60	262.89
Random search	289.38	257.48	303.78	279.68	334.72
(1+1)-EA + swap move	0.16	0.08	0.19	0.47	11.69
(1+1)-EA + swap + Max-Min seed	0.10	0.05	0.03	0.02	0.01
Max-Min heuristic	0.56	0.33	0.33	0.18	0.08
Min-Min heuristic	9.85	4.65	3.26	2.20	0.95

Table 3 · Runtime per run in seconds (s), mean over 10 runs

Algorithm (runtime in s)	500 tasks	1,000 tasks	1,500 tasks	2,000 tasks	5,000 tasks
DMO (original)	1.8	2.3	2.8	3.4	7.0
QI-DMO (quantum-inspired)	13.5	26.5	37.6	50.5	126.3
MRFO (original)	1.3	1.9	2.4	3.0	6.7
QI-MRFO (quantum-inspired)	13.2	26.0	38.3	51.1	134.2
QI-MRFO + swap move	14.6	28.2	40.9	55.6	139.0
QI-MRFO + swap + Max-Min seed	14.6	27.9	40.6	55.6	138.9
Genetic Algorithm (GA)	1.8	2.5	3.3	4.0	8.3
GA + swap move	2.4	3.4	4.1	5.0	10.1
PSO	1.3	2.0	2.7	3.3	8.0
Random search	0.5	0.7	1.0	1.3	2.9
(1+1)-EA + swap move	1.4	1.9	2.3	2.9	6.0
(1+1)-EA + swap + Max-Min seed	1.5	1.9	2.6	3.3	8.5
Max-Min heuristic	0.0	0.1	0.2	0.4	2.4
Min-Min heuristic	0.0	0.1	0.2	0.3	2.4

Measured on one shared machine with 4 runs in parallel, so the values compare the algorithms with each other rather than giving absolute speeds. The quantum-inspired methods keep a probability for every (task, VM) pair, so their time per evaluation grows with tasks × VMs.

Honest notes

- Without the swap move, QI-MRFO is behind the GA on average at 5 of 5 sizes; the swap move is what makes it competitive at this scale.
- A simple (1+1)-EA using the same swap move is better than the unseeded QI-MRFO + swap at 5 of 5 sizes, in every run (589.93 s vs 693.00 s at 5,000 tasks); seeded, the two are within 0.08 s of each other, and the (1+1)-EA takes 8.5 s per run against 138.9 s. For one-time batches the gain comes from the swap move and the seed, not from the swarm (as in the earlier held-out test H7).
- All methods used the same fixed budget of 20 000 evaluations. At 5,000 tasks this is small: the unseeded population methods end far from the bound (QI-MRFO + swap 31.2 % above it), so at thousands of tasks they need a larger budget or a heuristic seed.
- One problem per size and a synthetic workload; 10 runs per algorithm. Real traces are the next step.